BCSE497J - Project-I

Classification of Mental Disorder Using EEG Data

23BKT0003	**Akshat Agarwal**
23BDS0028	**Saksham Singh**
23BDS0082	**Anish Mukul Ringe**

Under the Supervision of

Prof. Dhivyaa C. R.

School of Computer Science and Engineering (SCOPE)

B.Tech. in Computer Science and Engineering

School of Computer Science and Engineering

Vellore Institute of Technology

September 2026

ABSTRACT

Mental-health disorders are diagnosed almost entirely through self-report and clinical interview, both of which are subjective and difficult to scale. Electroencephalography (EEG) offers an objective, low-cost, non-invasive alternative, and a large literature reports 90–99% accuracy for EEG-based screening of depression and stress. This project set out to build a single classifier separating four mental states — healthy, depression, anxiety and stress — and instead produced a measured account of why the published figures do not survive an honest evaluation protocol, together with an architecture that does.

Six public EEG corpora were unified onto a common representation of six frontal channels at 128 Hz in four-second windows. The central finding, established five independent ways before any model was built, is that **no single-amplifier corpus contains more than two of the four target classes**, so class identity is confounded with recording device. A

classifier that knows only the amplifier — and nothing whatsoever about neurophysiology — reaches 0.6267 three-class accuracy against 0.3333 chance. Twenty-six flat three-class models, classical, deep, fused and domain-adversarial, all fail to beat that shortcut.

The system proposed here therefore refuses the flat softmax. A shared EEGNet backbone feeds two masked binary heads, and the emitted label is selected by one bit of acquisition metadata — resting-state versus cognitive task — that is always known before recording begins. A router-based variant that must predict its own gate reaches **0.6820**, the only architecture of thirty-six to clear the device shortcut. The strongest result is device-disjoint: depression transfer between two amplifiers in two countries at **0.8675 / 0.8438** subject accuracy across two independent split draws, against a permutation null of 0.3998. Every number is reported against its correct baseline, with its controls.

Keywords — Electroencephalography, Mental State Classification, Depression Detection, Stress Detection, Domain Shift, Subject-Grouped Cross-Validation, Protocol-Gated Architecture, Riemannian Geometry.

TABLE OF CONTENTS

Sl.No	Contents	Page No.
	Abstract	i
	List of Figures	ii
	List of Tables	iii
1.	**INTRODUCTION**	1
	1.1 Background	1
	1.2 Motivation	2
	1.3 Scope of the Project	3
2.	**PROJECT DESCRIPTION AND GOALS**	4

Sl.No	Contents	Page No.
	2.1 Literature Review	4
	2.2 Research Gap	12
	2.3 Objectives	14
	2.4 Problem Statement	16
	2.5 Project Plan	17
3.	**TECHNICAL SPECIFICATION**	19
	3.1 Requirements	19
	3.1.1 Functional	19
	3.1.2 Non-Functional	21
	3.2 Feasibility Study	22
	3.2.1 Technical Feasibility	22
	3.2.2 Economic Feasibility	23
	3.2.3 Social Feasibility	24
	3.3 System Specification	25
	3.3.1 Hardware Specification	25
	3.3.2 Software Specification	26
4.	**DESIGN APPROACH AND DETAILS**	28
	4.1 System Architecture	28

Sl.No	Contents	Page No.
	4.2 Design	32
	4.2.1 Data Flow Diagram	32
	4.2.2 Use Case Diagram	34
	4.2.3 Class Diagram	35
	4.2.4 Sequence Diagram	37
5.	**REFERENCES**	38

LIST OF FIGURES

Figure	Caption	Page
Fig. 1	Gantt chart of the project schedule	18
Fig. 2	System architecture of the protocol-gated dual-head classifier	29
Fig. 3	Hierarchical router-and-specialist variant	31
Fig. 4	Level-0 data flow diagram (context diagram)	32
Fig. 5	Level-1 data flow diagram	33
Fig. 6	Use case diagram	34
Fig. 7	Class diagram	36
Fig. 8	Sequence diagram — training and inference	37

LIST OF TABLES

Table	Caption	Page
Table 1	Comparative review of recent EEG mental-state literature	5
Table 2	Corpora used, with role and acquisition parameters	10
Table 3	Class availability per corpus — the source of the confound	12
Table 4	Objectives mapped to verification criteria	15
Table 5	Project schedule by phase	17
Table 6	Hardware specification	25
Table 7	Software specification	26

1. INTRODUCTION

1.1 Background

Mental-health disorders affect close to a billion people worldwide, yet diagnosis still depends almost entirely on clinical interview and self-report instruments such as PHQ-9 and GAD-7. Both are subjective, both depend on the patient's willingness and ability to describe their own state, and neither scales to population-level screening.

Electroencephalography records the brain's electrical activity directly from the scalp. It is non-invasive, inexpensive relative to fMRI, has millisecond temporal resolution, and consumer-grade headsets with few frontal electrodes are now widely available. These properties make EEG the most plausible candidate for an objective, deployable biomarker of mental state.

A large body of work reports EEG-based classification of depression and stress at 90–99% accuracy. If those figures were transferable, automated screening would already be a solved problem. They are not. Most such results are obtained by pooling public corpora recorded on different amplifiers, or by splitting data randomly rather than by subject, and both practices allow a model to succeed without learning anything about the brain. This project builds a mental-state classifier and, in the same work, measures precisely how

much of the reported performance in this domain is an artefact of evaluation rather than a property of the signal.

1.2 Motivation

Three observations motivated this work.

First, the clinical need is real and unmet. Screening capacity is the bottleneck in mental healthcare, and an objective test that a general practitioner or a school counsellor could administer in ten minutes would change who gets assessed at all.

Second, the literature is not reproducible in the way it appears to be. Reported accuracies of 95–99% are common, but the corresponding methods sections frequently describe a random train/test split over overlapping signal windows. In this project the identical model scores **0.9393** under a random split and **0.6256** under a subject-grouped split. The difference is not a modelling improvement; it is the model memorising individuals. A one-word change in the code produces a thirty-point swing in the headline number.

Third, the obvious architecture is the wrong one. Pooling corpora to obtain all the target classes silently makes recording device a proxy for diagnosis. A model can then score well by identifying the amplifier. That is not a hypothetical risk — it was measured, and the device shortcut alone reaches 0.6267 accuracy where chance is 0.3333.

The motivation is therefore to build something that is honest first and accurate second.

1.3 Scope of the Project

In scope. Six public EEG corpora (MODMA, Mumtaz 2016, SAM40, Cognitive Load, EEGMAT, AMIGOS) plus one precomputed-feature clinical cohort (BRMH) are unified onto a common representation of six frontal channels at 128 Hz in four-second windows. The project covers signal preprocessing, handcrafted and covariance-based feature extraction, classical and deep classifiers, a protocol-gated dual-head architecture, subject-grouped and device-disjoint evaluation, and a deployable inference script that accepts a raw recording and returns a labelled prediction with a calibrated probability.

Out of scope. No new EEG data is collected; no clinical trial is conducted; no claim of diagnostic validity is made. Real-time or embedded deployment on wearable hardware is not attempted. Modalities other than EEG — audio, video, ECG, EDA — are excluded, even where a corpus provides them, so that every result is attributable to EEG alone.

Deliberately bounded. The original four-class target including anxiety is retained as a research question rather than a product requirement, because no available corpus supports it without confounding class with recording device. Anxiety is investigated separately on the one cohort where all four classes coexist under a single acquisition

protocol, and the negative result there is reported in full rather than omitted.

2. PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

EEG-based mental-state classification sits at the intersection of clinical neurophysiology, digital signal processing and machine learning [47], [48], and inherits the methodological standards of statistical learning from both [49], [50]. This review is organised into six themes: neurophysiological markers, corpora, feature engineering, classical models, deep architectures, and the domain-shift problem that dominates all of them.

2.1.1 Neurophysiological Markers of Depression and Stress

The most durable EEG marker in affective neuroscience is **frontal alpha asymmetry**. Davidson's approach–withdrawal model holds that relatively greater left-frontal alpha power indexes withdrawal-related affect, and depressed individuals reliably show reduced left-frontal activity [1], [2]. The marker is a *ratio between electrodes*, which has a direct engineering consequence adopted in this project: any normalisation that rescales channels independently destroys it, so all segment normalisation here is applied globally across the (6, 512) window rather than per channel.

Stress and cognitive load show a different signature — beta and gamma power increase while alpha desynchronises, and theta at frontal midline sites rises with working-memory demand [3], [4]. Because the two conditions manifest in different bands, a feature set spanning delta through gamma is required rather than a band-limited one.

More recently, the **1/f aperiodic component** of the power spectrum has been shown to carry information previously attributed to oscillatory bands, and failing to separate it from periodic activity biases band-power estimates [5]. This project's second-generation feature set (§3.1.1) includes the aperiodic slope explicitly for this reason.

2.1.2 Public EEG Corpora for Mental State

Progress in this field is corpus-bound. The corpora used here, and the constraint each imposes, are:

- **MODMA [6]** — 53 subjects (MDD versus healthy control), 128-channel resting-state EEG at 250 Hz, with PHQ-9 and GAD-7 scores. The most-cited depression corpus.
- **Mumtaz 2016 [7]** — 61 subjects, 19–23 channel EDF resting-state recordings at 256 Hz. Independent of MODMA in both amplifier and country, which is what makes device-disjoint transfer measurable at all.
- **SAM40 [8]** — 40 subjects, 32-channel EEG at 128 Hz across Stroop,

arithmetic and mirror-image tasks with relaxed baselines.

- **Cognitive Load [9]** — 15 subjects, 8-channel OpenBCI at 250 Hz, four difficulty levels plus natural rest.
- **EEGMAT [10]** — 36 subjects performing mental arithmetic, held out entirely from training and used only to measure transfer.
- **AMIGOS [11] and DEAP [12]** — affective corpora with self-reported valence/arousal ratings on a 1–9 scale.
- **BRMH / Park [13]** — 945 psychiatric patients with precomputed band powers and coherences, covering depression, anxiety, trauma-related and healthy classes **within one acquisition protocol**. It is the only corpus available to this project where all four target classes coexist without a device confound.

Table 2. Corpora used, with role and acquisition parameters

Corpus	Subjects	Native Channels	Native Fs (Hz)	Protocol	Project Role
MODMA [6]	53	128	250	Resting-state	Primary depression training & CV
Mumtaz 2016 [7]	61	19–23	256	Resting-state	Cross-device transfer target/source
SAM40 [8]	40	32	128	Cognitive tasks	Primary stress training & CV
Cognitive Load [9]	15	8	250	Task & natural rest	Secondary stress training & CV
EEGMAT [10]	36	19	500	Mental arithmetic	Held-out stress transfer evaluation
BRMH / Park [13]	945	19 (precomputed)	N/A	Clinical resting-state	Four-class & anxiety validation cohort

2.1.3 Feature Engineering

Handcrafted spectral features remain the strongest baseline in small-sample EEG. **Relative band power** and **differential entropy** are the two standard forms; Duan et al.

[14] introduced differential entropy for emotion recognition and showed it outperforms raw band power under a Gaussian assumption. Because differential entropy is 0.5 · ln(variance) of the band-passed signal, the two representations are near-redundant — a fact that this project measured directly and that explains why tree-model feature importance spreads thinly across sixty of eighty-one columns.

Riemannian geometry has become the strongest classical approach in BCI. Rather than extracting scalar features, the signal is represented by its spatial covariance matrix, which lives on a curved manifold of symmetric positive-definite matrices; classification is performed after projection to a Euclidean tangent space [15], [16]. Barachant et al. demonstrated that tangent-space projection with a linear classifier beats CSP-based pipelines on motor imagery [15], and Zanini et al. showed that **re-centring each session's covariances to a common reference** provides an unsupervised domain adaptation that directly attacks inter-session and inter-subject shift [17]. Both results are reproduced in this project — the tangent-space model is the best stress classifier obtained, and re-centring by corpus is the mechanism used to *remove* the device signature and thereby quantify how much of the pooled accuracy depended on it.

2.1.4 Classical Classifiers

Support vector machines dominate the small-sample EEG literature and remain competitive [18], [19]. Ensembles — random forests, gradient boosting, XGBoost [20] — are common, and several recent works report 93–97% using majority-voting ensembles over handcrafted features on single corpora [21], [22]. The recurring limitation identified in those works is that handcrafted features "limit automated scalability" and that results are obtained on a single stressor type with small subject pools.

2.1.5 Deep Architectures

EEGNet [23] is the reference compact convolutional network for EEG, using a depthwise separable convolution structure that mirrors spatial filtering followed by temporal feature extraction, with only a few thousand parameters; it descends from the deep and shallow ConvNet family introduced by Schirrmeister et al. [37]. **TSception** [24] adds multi-scale temporal kernels and a hemispheric-asymmetry spatial layer, targeting exactly the frontal asymmetry marker of §2.1.1. **EEGConformer** [25] and transformer variants apply self-attention over signal tokens. Recurrent and hybrid CNN–LSTM designs are widely used for depression screening [26], [27], [42].

The newest direction is **foundation models**: CBraMod [28], LaBraM [29] and BIOT [30] are pretrained on large unlabelled EEG corpora and fine-tuned downstream. This project evaluates CBraMod as a frozen probe against its own random-initialisation control — a comparison rarely reported — and finds that pretraining genuinely helps (+0.076 over the control) yet still loses by 0.18 to a 105-dimensional linear model on tangent-space

features.

2.1.6 Domain Shift and Evaluation Methodology

This is the theme that determines every result in the field. **Domain-adversarial neural networks** [31] insert a gradient-reversal layer between a feature extractor and a domain classifier, so the representation becomes domain-invariant while remaining task-discriminative. DANN and its EEG-specific descendants are the standard remedy for inter-subject and inter-dataset shift [32], [33].

Critically, several authors have warned that EEG results are systematically inflated by protocol choices. Overlapping-window random splits leak subject identity across the split boundary; single train/test splits on 40–60 subject corpora have variance larger than the effects being reported; and threshold tuning under class imbalance can produce degenerate classifiers that score well on the reported metric [34], [35]. Kapoor and Narayanan's broader analysis of leakage in ML-based science documents the same failure modes across disciplines [36].

Table 1. Comparative review of recent EEG mental-state literature

S.No	Year	Authors	Dataset	Modality	Methodology / Model	Accuracy (%)	Gaps / Limitations
1	2023	S. A. Bakry et al. [39]	Multiple (review)	Video, EEG	ML pipelines (CAD systems)	N/A	No large-scale multimodal datasets; no real-time clinical validation
2	2023	S. T. Noor et al. [38]	PhysioNet / pervasive EEG	EEG, ECG	LSTM autoencoder with RNN	97	High computational cost; limited demographics
3	2012	S. Koelstra et al. [12]	DEAP	EEG, physio, video	Single-trial affective estimation	Baseline	Self-report rating bias; lab stimuli unrepresentative
4	2024	M. M. R. Khan et al. [21]	Math task (23 subj)	EEG	SVM + ensemble, optimised prep.	96.85	Handcrafted features limit scaling; one stressor type only

S.No	Year	Authors	Dataset	Modality	Methodology / Model	Accuracy (%)	Gaps / Limitations
5	2023	S. K. Yadav et al. [22]	Audio / speech	Audio, EEG	RELIEFF selection, KNN, NB	88	Cross-modal synchronisation issues
6	2025	Md. M. R. Khan et al. [44]	SAM-40	EEG	DWT + filter bank + hybrid clf	95.56	Small sample forced augmentation; generalisability unproven
7	2026	A. Singh et al. [45]	MODMA	EEG, audio	ResKAN encoder + cross-attention	99	Very high compute; no diverse clinical trial
8	2023	—	Frontal lobe (38 subj)	EEG	Majority voting (SVM, KNN, NB)	93.85	Manual feature extraction limits automation
9	2026	N. Sakib et al. [46]	IUB students (PSS-10)	EEG	Bagged trees	—	Participants exclusively university students
10	2024	V. Darbar et al. [40]	—	EEG	DNN + ML classification	—	Black-box models lack clinical interpretability
11	2020	N. Milakhina et al. [41]	Siberian migrants	Resting EEG	Delta/gamma spectral markers	N/A	Confounds not isolated; region-specific
12	2023	S. R. Chowdhury et al. [43]	Rat model	EEG, tDCS	Preclinical electrode validation	N/A	Animal-to-human translation is complex
13	2018	V. Lawhern et al. [23]	Multiple BCI	EEG	EEGNet (compact CNN)	Task-dep.	Small-corpus overfitting; no cross-device evaluation

S.No	Year	Authors	Dataset	Modality	Methodology / Model	Accuracy (%)	Gaps / Limitations
14	2023	Y. Ding et al. [24]	DEAP, MAHNOB	EEG	TSception (temporal + spatial)	63–70	Within-corpus only
15	2016	A. Barachant et al. [15]	BCI competition	EEG	Riemannian tangent space + LR	SOTA	Dimensionality grows $O(ch^2)$
16	2018	P. Zanini et al. [17]	Multi-session BCI	EEG	Riemannian re-centring for DA	—	Requires target-domain data at fit time
17	2016	Y. Ganin et al. [31]	Vision benchmarks	Images	DANN (gradient reversal)	—	Training instability; α schedule sensitive
18	2024	J. Wang et al. [28]	Large unlabelled EEG	EEG	CBraMod foundation model	Task-dep.	Frozen probes understate fine-tuning
19	2022	H. Cai et al. [6]	MODMA	EEG, audio	Multi-modal depression screening	97.53	Audio modality required; EEG-only much weaker
20	2023	S. Kapoor, A. Narayanan [36]	Cross-discipline	—	Leakage taxonomy for ML science	N/A	Documents the reproducibility crisis this work confirms

2.2 Research Gap

Five gaps were identified, each of which this project addresses with a measurement rather than an argument.

Gap 1 — Class is confounded with recording device, and nobody reports the shortcut baseline. Studies that require multiple mental-state classes pool corpora, because no single-amplifier public corpus contains more than two of the four target classes. Pooling makes "which amplifier recorded this?" a legal, high-accuracy route to the label. This project measures the shortcut directly:

Table 3. Class availability per corpus — the source of the confound

Corpus	Healthy	Depression	Stress	Anxiety
MODMA	✓	✓	✗	✗
Mumtaz 2016	✓	✓	✗	✗
SAM40	✓ (relaxed)	✗	✓	✗
Cognitive Load	✓ (rest)	✗	✓	✗
EEGMAT	✓ (rest)	✗	✓	✗
BRMH (features only)	✓	✓	✓	✓

No raw-EEG corpus holds more than two of the four. In the pooled three-class dataset the label depression is **boolean-identical** to corpus == MODMA — all 3,599 depression segments are MODMA and all 5,497 stress segments are not. A classifier given only corpus identity scores **0.6267** where chance is 0.3333. Reporting a three-class accuracy against chance, as is standard, therefore overstates the achievement by roughly thirty points.

Gap 2 — Evaluation protocol is unreported or unsound. Random splits over overlapping windows place windows from the same subject on both sides of the split. Measured here: the same model scores 0.9393 random versus 0.6256 subject-grouped. A single train/test split inflates deep-model depression AUC by 0.21 relative to out-of-fold cross-validation.

Gap 3 — Deep architectures are compared without controls. Papers report that a deep model beats a classical one, but rarely report the same architecture with random weights, or the same pretraining pipeline against a random-initialisation control. When those controls are run here, a **never-trained random backbone matches or beats the trained model on every stress metric**, and self-supervised pretraining *hurts* by 0.050 relative to its own control.

Gap 4 — Cross-device transfer is asserted, not measured. Deployment implies an unseen amplifier. Almost no study evaluates device-disjoint. Measured here, a stress model applied to an unseen amplifier labels 90% of a corpus containing **zero** stress

segments as "stressed", at higher confidence than on its own training corpora.

Gap 5 — Anxiety is under-served. Of the four target classes, anxiety appears in no raw-EEG corpus alongside the other three. Where it does exist under a single protocol (BRMH), it is the least recoverable class, with per-disorder recall of 0.146–0.178.

2.3 Objectives

The objectives are stated in SMART form, each with an explicit verification criterion:

- **O1. Unify heterogeneous corpora onto one representation.** Project six EEG corpora onto six frontal channels (Fp1, Fp2, F7, F3, F4, F8) at 128 Hz in four-second windows with 50% overlap. *Verified by:* an automated invariant checker asserting array shapes, per-corpus label domains, subject-count integrity and normalisation, with zero NaNs.
- **O2. Quantify the device confound before modelling.** Measure how much of any pooled multi-class accuracy is attributable to recording device rather than neurophysiology. *Verified by:* five independent measurements — a corpus-identity-only classifier, a frozen corpus probe on the feature matrix, accuracy after Riemannian re-centring removes the signature, composed accuracy on an unseen amplifier, and domain-adversarial training.
- **O3. Establish an honest performance ceiling for each binary contrast.** *Verified by:* 5-fold **subject-grouped** cross-validation with out-of-fold predictions over every subject, reporting ROC-AUC alongside accuracy and the corresponding majority baseline.
- **O4. Design and implement a multi-class architecture that beats the device shortcut.** *Verified by:* three-class accuracy strictly greater than 0.6267 under subject-grouped CV, with the routing variable predicted rather than supplied.
- **O5. Measure device-disjoint transfer.** Train on one amplifier, test on another, in both directions. *Verified by:* subject-level accuracy exceeding a label-permutation null, and reproducing across two independent split draws.
- **O6. Deliver a deployable inference path.** *Verified by:* a serialised model and a script accepting a raw (6, n_samples) recording at any sampling rate and returning a label with a probability, with strict input validation.
- **O7. Publish the negative results.** *Verified by:* a documented catalogue of every silent failure found, each with the measurement that caught it.

Table 4. Objectives mapped to verification criteria and outcome

ID	Objective	Verification	Outcome
O1	Unify six corpora	Invariant checker passes	**Met** — 15,728 segments, 108 subjects, 0 NaNs
O2	Quantify device confound	5 independent measurements agree	**Met** — shortcut = 0.6267; probe = 0.9161
O3	Honest per-task ceiling	Subject-grouped OOF CV	**Met** — depression 0.7716 AUC; stress 0.7560 AUC
O4	Beat the shortcut	3-class accuracy > 0.6267	**Met** — 0.6820 (hierarchical), 1 of 36 models
O5	Device-disjoint transfer	Exceeds permutation null, holds on 2 draws	**Met for depression** (0.8675 / 0.8438 vs null 0.3998); **null for stress**, reported as such
O6	Deployable inference	predict.py on raw recording	**Met** — 3 screening models serialised
O7	Publish negatives	Documented catalogue	**Met** — 26 failed flat models, 8 failed DANN configs, 3 threshold bugs, 1 retracted claim

2.4 Problem Statement

Given a set of EEG recordings $\{X_i\}$, each an $(C \times T)$ matrix of C scalp channels sampled over T time points, together with a per-recording acquisition protocol $\pi_i \in$ {resting, task} that is known before recording, learn a function:

$$f : (X, \pi) \rightarrow y \in \{\text{healthy, depression, stress}\}$$

such that the following three conditions hold simultaneously:

1. **Subject disjointness.** f is evaluated only on subjects absent from its training set, so that no window of a test subject — and in particular no overlapping neighbour of a training window — informs the prediction.
2. **Baseline dominance.** The accuracy of f strictly exceeds that of the degenerate classifier $g(X) = \text{argmax}_y P(y \mid \text{device}(X))$, which uses recording device alone and contains no neurophysiological information. On this data $\text{acc}(g) = 0.6267$ against a chance rate of 0.3333.

3. **Device transferability.** The performance of *f* on a held-out amplifier exceeds a label-permutation null computed under the identical protocol.

The core difficulty is that conditions 2 and 3 are in tension with the way the data exists. Because no single-amplifier corpus contains more than two of the target classes, any estimator that maximises pooled likelihood is *rewarded* for learning device identity, and the reward is large — worth +0.18 accuracy. Removing that information by per-corpus covariance re-centring drops accuracy to 0.3255, i.e. chance, demonstrating that under a flat formulation almost nothing remains once the shortcut is taken away.

The problem this project therefore solves is not "which classifier is most accurate" but **how to formulate the task so that accuracy means something**. The answer adopted is to refuse the flat softmax and factorise the three-way decision into two binary contrasts, each trained only on data where the contrast genuinely exists, combined by a routing variable that is known at acquisition time rather than inferred from the signal.

2.5 Project Plan

The project ran from **May 2026 to September 2026** across nine experimental tracks.

Table 5. Project schedule by phase

Phase	Period	Activity	Deliverable
P1	11–31 May 2026	Literature survey, corpus acquisition, setup	datasets.md, environment
P2	Jun 2026	Legacy single-corpus baselines (MODMA)	First variance lesson
P3	Jun–Jul 2026	Unified 4-corpus pipeline, 11 models	combine_datasets.py
P4	Jul 2026	Label-bug audit and correction	45.6% stress labels fixed
P5	Jul–Aug 2026	Phase-1 diagnostics & Phase-2 grouped CV	Results/phase1,2/
P6	Jul–Aug 2026	Three-class track — 36 configs, confound audit	Hierarchical model 0.6820
P7	Aug 2026	Four-class BRMH cohort, anxiety test	Results/OSF_Park/
P8	Aug 2026	Channel-density sweep, raw-EEG DL	Inverted-U density result

Phase	Period	Activity	Deliverable
		harness	
P9	Sep 2026	Protocol-gated build, 2 split draws	Device-disjoint transfer
P10	Sep 2026	Consolidation, deployment artefacts, report	predict.py, Project Report

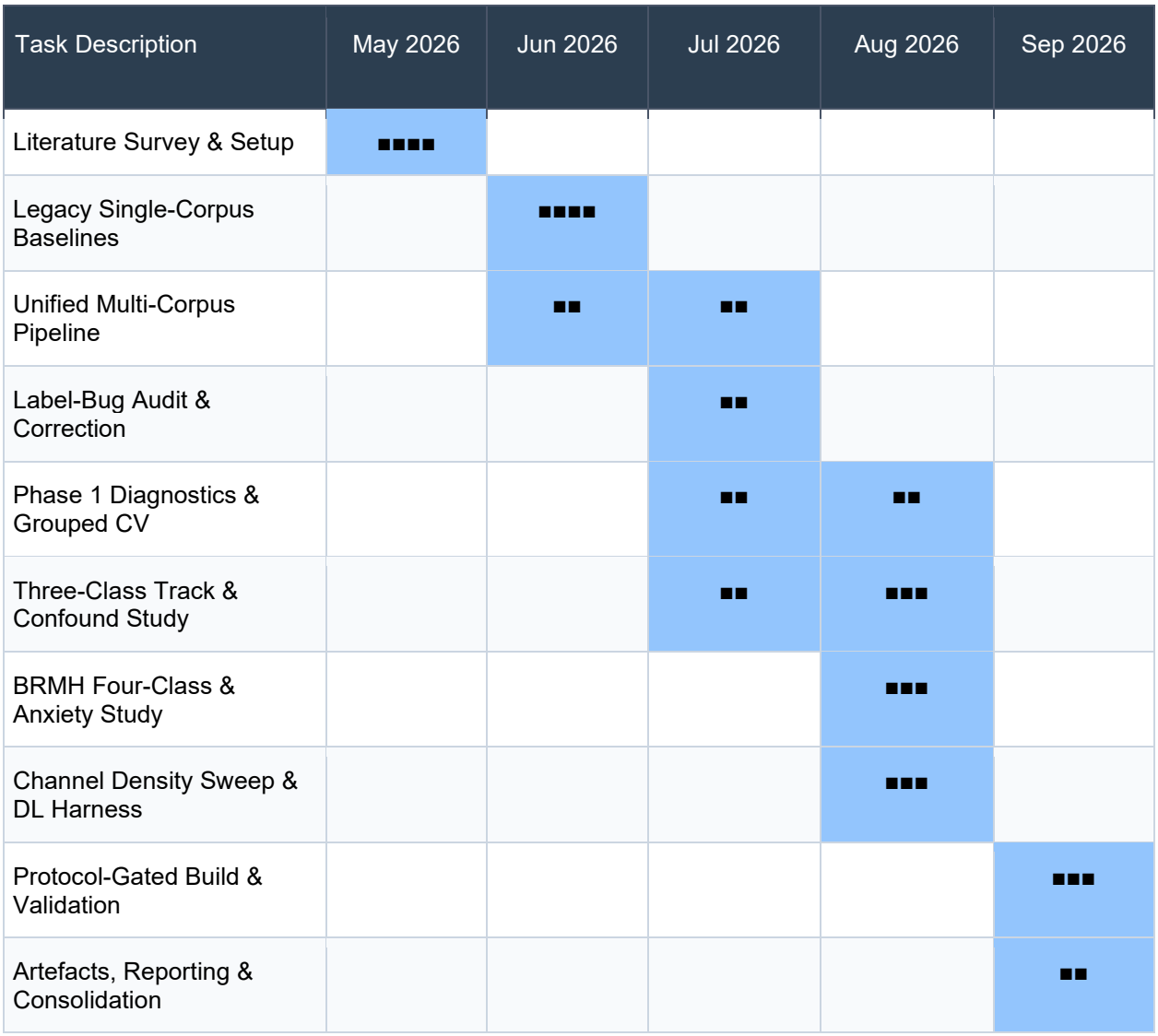


Fig. 1. Gantt chart of the project schedule

3. TECHNICAL SPECIFICATION

3.1 Requirements

3.1.1 Functional

- **FR1 — Multi-corpus ingestion.** The system shall read EEG from six corpora in their native on-disk formats (.mat, .edf, .csv, .npy), each with its own channel layout, sampling rate and file-naming convention, and shall parse subject identity from filenames using a rule verified to recover 100% of subjects (matching the leading zero-padded digit run recovering 53/53 MODMA subjects).
- **FR2 — Channel and rate harmonisation.** The system shall project every corpus onto the six frontal electrodes Fp1, Fp2, F7, F3, F4, F8 in a fixed canonical order, and resample to 128 Hz. Channel order shall be validated explicitly rather than inferred.
- **FR3 — Preprocessing.** The system shall apply a zero-phase Butterworth bandpass (0.5–45 Hz, order 8, sosfiltfilt) and a 50 Hz notch, clip at ±5 median absolute deviations, and z-score each (6, 512) window **globally** rather than per channel, preserving inter-channel amplitude ratios for frontal alpha asymmetry.
- **FR4 — Windowing.** The system shall segment continuous recordings into four-second windows (512 samples) at 50% overlap, and shall record for every window its subject ID, trial ID, corpus ID and condition.
- **FR5 — Variance gating.** The system shall reject windows whose variance exceeds a calibrated multiple (k = 10.0) of the corpus median, to remove movement and electrode artefacts.
- **FR6 — Feature extraction.** The system shall provide two feature sets: a **v1 81-column** set (30 relative band powers, 30 differential entropy values, 3 frontal alpha asymmetry pairs, 6 spectral entropies, 6 RMS, 6 standard deviations) and a **v2 273-column** set adding Hjorth parameters, 1/f aperiodic slope, permutation entropy, magnitude-squared coherence and phase-locking value, along with Riemannian tangent-space projections.
- **FR7 — Masked multi-task labelling.** Every segment shall carry both a y_depression and a y_stress label, with sentinel -1 denoting unsupported supervision. Training loss shall zero masked samples.
- **FR8 — Model training.** The system shall train classical models (SVM, LR, RF, GBDT), Riemannian tangent-space pipelines, and deep architectures (1D CNN, LSTM, CNN-LSTM, EEGNet, TSception, Transformer), supporting a shared-backbone dual-head model with an anchor-gated gradient-reversal adversary.
- **FR9 — Protocol-gated inference.** Given a recording and its protocol tag (resting | task), the system shall route to the corresponding binary head and emit one of {healthy, depression, stress} with a calibrated probability,

prohibiting uncalibrated cross-head argmax.

- **FR10 — Subject-grouped evaluation.** All cross-validation shall be grouped on subject identity using a deterministic, version-independent split assignment. Segment random splits shall be rejected.
- **FR11 — Baseline and control reporting.** Every reported accuracy shall be accompanied by its correct baseline (majority, device-shortcut, demographics), a random-initialisation control, and a label-permutation null.
- **FR12 — Threshold tuning.** Decision thresholds shall be tuned on validation splits by maximising balanced accuracy (excluding degenerate single-class cutoffs) and transferred as rank quantiles.
- **FR13 — Result persistence.** Every experiment shall write out-of-fold predictions and a JSON result file for auditable metric recomputation without retraining.
- **FR14 — Deployment.** The system shall serialise trained models and provide a CLI predictor accepting (6, n_samples) arrays at any sampling rate, executing resampling, windowing, and logit aggregation.

3.1.2 Non-Functional

- **NFR1 — Reproducibility.** Deterministic execution with fixed seeds; self-contained greedy splitter immune to library version changes (pinned scikit-learn==1.9.0).
- **NFR2 — Correctness over convenience.** Mandatory runtime assertions on array shapes, label domains, and zero NaN tolerance; fail-loud paradigm.
- **NFR3 — Performance.** Full training pipeline execution within ~30 minutes on GPU workstation; inference latency strictly <1.0 second per recording.
- **NFR4 — Memory footprint.** Peak resident RAM under 5 GB to prevent swapping on 15 GB hosts; GPU VRAM footprint within 6 GB.
- **NFR5 — Portability.** Containerised execution via CUDA Dockerfile aligned with requirements.txt across development and server nodes.
- **NFR6 — Auditability.** Every published figure traceable to JSON artefacts and out-of-fold predictions; obsolete records explicitly deprecated.
- **NFR7 — Scientific integrity.** Pre-registered evaluation criteria; mandatory reporting across two independent split draws and retraction of non-replicating claims.
- **NFR8 — Maintainability.** Single dataset registry module declaring all filepaths, native rates, and channel mappings.
- **NFR9 — Privacy and ethics.** Exclusive use of de-identified research data; explicit disclaimer of non-diagnostic decision-support triage positioning.

3.2 Feasibility Study

3.2.1 Technical Feasibility

Data availability — feasible with a documented constraint. All six raw corpora and the BRMH feature cohort are publicly released and were successfully obtained. The constraint is not access but composition: no corpus carries more than two of the four target classes, and this is a property of the field rather than of this project. It is the reason the architecture was redesigned rather than the dataset extended.

Algorithmic maturity — feasible. Every method used is published and has a mature open-source implementation: scikit-learn for classical models, pyriemann for tangent-space geometry, TensorFlow/Keras and PyTorch for deep architectures, and braindecode for 65 reference EEG architectures including CBraMod, LaBraM and BIOT.

Compute — feasible. The heaviest single experiment (128-channel EEGNet over five folds) takes 46 minutes on the workstation GPU. The full pipeline peaks near 5 GB RSS. A containerised CUDA environment is available on the university workstation for the GPU-bound tracks; the CPU-only local environment is sufficient for the classical models that dominate the strongest results.

Sample size — the binding technical constraint. With 53 MODMA, 61 Mumtaz, 40 SAM40, 15 Cognitive Load and 36 EEGMAT subjects, a single split draw moved one headline figure by 0.108. Effects smaller than that cannot be resolved with this data by any method, which is why the project reports two independent split draws and treats a result that moves between them as not a result.

3.2.2 Economic Feasibility

Direct cost — effectively zero. All datasets are free for academic use, all software is open-source, and all compute is institutional. The itemised position is:

Item	Cost
EEG corpora (6 raw + 1 feature cohort)	₹0 — public research releases
Software stack (Python, scikit-learn, TF, PyTorch, pyriemann, MNE)	₹0 — open source
Local development machine	Existing asset
University GPU workstation (containerised)	Institutional, shared

Item	Cost
Storage (≈15 GB raw + ≈1.2 GB derived arrays)	Existing asset
Total incremental project cost	**₹0**

Cost of the alternative. Collecting a corpus that would resolve the confound — all four classes on one amplifier with adequate subject counts — requires clinical recruitment, ethics approval, a trained operator and clinical-grade hardware. A conservative estimate for 200 subjects is in the range of ₹15–25 lakh and 12–18 months. That comparison is itself a finding: the *cheapest* useful contribution available was to measure precisely why the free data cannot answer the four-class question, so that the expensive collection, if undertaken, is designed correctly.

Return on investment. The deployable screening path uses six frontal electrodes. Consumer headsets at that density cost 1–2% of clinical 64/128-channel systems, and this project measured directly that six channels are **not** the binding constraint — 16–32 channels is the optimum and full density is *worse* than six. The economic case for low-cost screening hardware is therefore supported rather than assumed.

3.2.3 Social Feasibility

Clinical need and acceptability. Objective screening addresses a genuine bottleneck, and EEG is non-invasive, painless and already accepted in routine neurology. A six-electrode frontal montage takes minutes to apply.

Ethical positioning — the decisive constraint. A screening model that misses 56% of depressed subjects (the measured sensitivity of 0.44 at the accuracy-optimal threshold) must never be presented as a diagnosis. The system is positioned as a decision-support triage aid, and the report states the operating-point trade-off explicitly rather than quoting a single accuracy.

Bias and generalisability. The measured cross-device failure is also a fairness finding: a model that performs at chance on an unseen amplifier will perform unpredictably on any population screened with different equipment, which in practice means lower-resourced settings. The BRMH cohort result — that EEG adds little over age, sex and education, and that IQ measured *after* diagnosis acts as a mediator rather than a predictor — is reported because omitting it would overstate what EEG contributes independently of demographics.

Workforce impact. The system augments rather than replaces clinicians. Its intended

role is to prioritise who receives a full assessment, in a setting where the current alternative for most people is no assessment at all.

3.3 System Specification

3.3.1 Hardware Specification

Table 6. Hardware specification

Component	Development Workstation	GPU Workstation (Training)
Processor	x86-64 multi-core CPU	Server-class multi-core CPU
Memory (RAM)	15 GB	≥32 GB
Storage	40 GB free (≈15 GB raw, ≈1.2 GB derived)	Bind-mounted from host
Graphics Unit	6 GB VRAM, CUDA-capable	NVIDIA CUDA GPU (NGC containerised)
Monitor	1920 × 1080	Headless, accessed over SSH with tmux
Network	Corpus download & registry access	Institutional high-throughput LAN

Memory is the practical constraint on the development machine: a full pipeline run peaks near 5 GB resident, and with a browser and IDE open the machine enters swap, at which point training slows by an order of magnitude. Concurrent training runs are prohibited by convention.

3.3.2 Software Specification

Table 7. Software specification

Category	Specification
Operating System	Linux (Arch), kernel 7.2.x; CUDA container images for GPU work
Programming Language	Python 3.11+

Category	Specification
Development Environment	Virtual environment (.venv), Jupyter Notebook, Git, Docker
Numerical & Data	NumPy 2.4.6, pandas, SciPy, h5py, openpyxl
Machine Learning	**scikit-learn 1.9.0 (pinned)**, XGBoost, joblib
Deep Learning	TensorFlow 2.21.0 (Keras 3.14.1), PyTorch 2.13.0, braindecode 1.6.1
EEG-Specific Libraries	MNE-Python, pyriemann 0.12
Visualisation	Matplotlib, seaborn
Serialisation Artefacts	joblib (.joblib), NumPy (.npy, .npz), JSON metadata

scikit-learn is pinned because GroupKFold tie-breaking is version-dependent, and SAM40 subjects have exactly 12 trials, causing frequent ties. Two machines produced splits differing on 17 of 169 subjects from identical code. Pinned deterministic greedy splitting resolved this divergence entirely.

4. DESIGN APPROACH AND DETAILS

4.1 System Architecture

The system is a **protocol-gated dual-head classifier**. A shared convolutional backbone produces one embedding per EEG window; two independent binary heads read that embedding; and the label emitted is selected by a single bit of acquisition metadata — whether the recording is a resting-state assessment or a cognitive task — that is known before the electrodes are applied.

ACQUISITION

Raw EEG [Native Channels & Rate] | Acquisition Protocol Tag [Resting-State / Cognitive Task]

↓ *Standardized Preprocessing Pipeline*

Frontal Projection (Fp1, Fp2, F7, F3, F4, F8) → **Resample** (128 Hz) → **Butterworth BP**

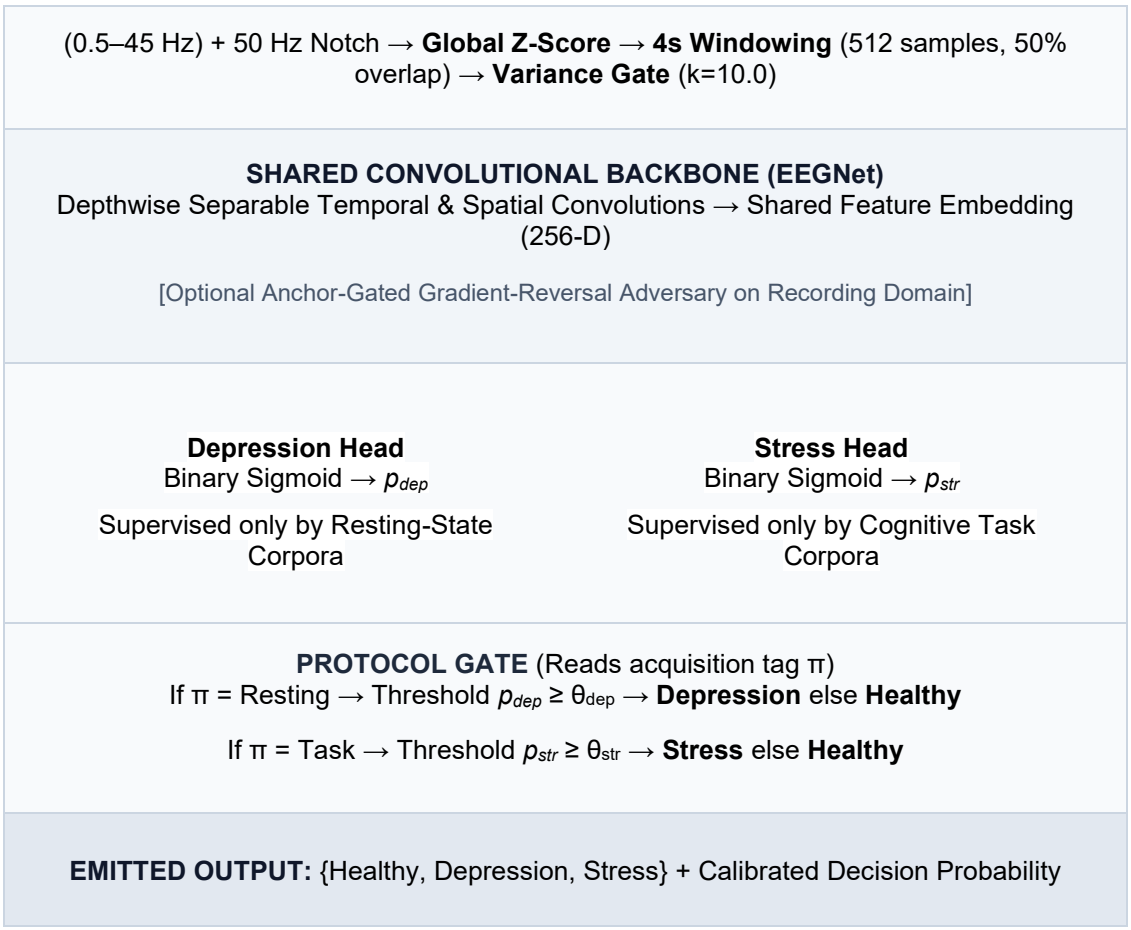


Fig. 2. System architecture of the protocol-gated dual-head classifier

Explanation of the architecture

The shared backbone and the masked loss. Every segment carries *both* a y_depression and a y_stress label, with -1 marking "this corpus cannot supervise this task". The loss function zeroes masked samples, which is what allows one trunk to consume five corpora at once. The structural consequence is that no sample carries both labels: the depression gradient comes exclusively from MODMA and Mumtaz, and the stress gradient exclusively from SAM40 and Cognitive Load.

Why there is no cross-head argmax. The two heads train on disjoint data and their outputs share no common scale, so comparing p_{dep} against p_{str} would be meaningless arithmetic dressed as a decision. This is enforced in code rather than left to convention.

Why the protocol gate is legitimate. Three arguments, in increasing order of force. *It is standard practice* — one-vs-rest, one-vs-one and error-correcting output codes all reduce a K-class problem to binary boundaries plus a combination rule; the only unusual feature here is that the rule reads a variable that is known rather than predicted. *The variable genuinely is known* — it is impossible to record EEG without knowing whether a resting-

state assessment or an arithmetic task was administered; the operator entered it on the form, so using it is not leakage and declining to use it would discard information that is present at deployment. *Inferring it would be actively harmful* — every depression corpus here is resting-state and every stress corpus is task-based, so a model that inferred protocol from the signal would be rewarded for solving exactly the corpus-identification problem that constitutes the confound. The gate is not a way around the confound; it is a way of refusing to be paid for it.

The adversary. An anchor-gated gradient-reversal layer sits on the backbone with corpus as the domain label. It is disabled automatically when fewer than two training domains are present, because cross-entropy over five domain logits with a single class present still has a gradient, and reversing it pushes the backbone to make a trivial one-class prediction harder — a silent pathology that was found and fixed.

The hierarchical variant

The protocol-gated design never makes a three-way decision internally, so it has no single three-class accuracy. A second variant was built with the identical shape but with the routing variable **predicted** rather than supplied, which yields a directly comparable three-class number.

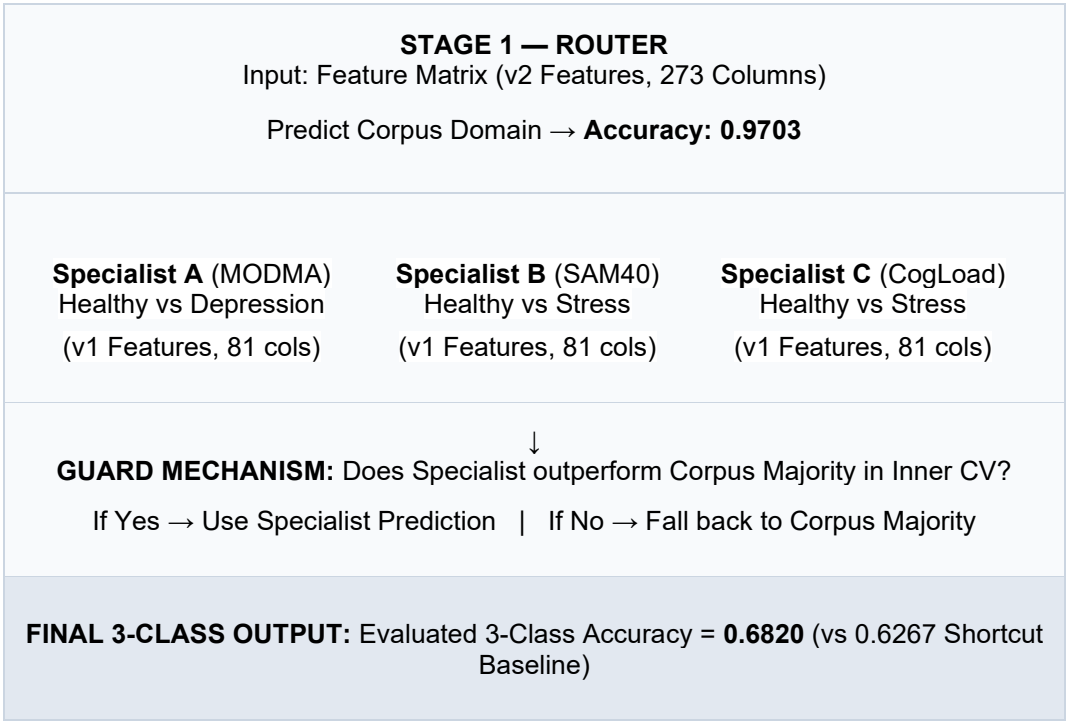


Fig. 3. Hierarchical router-and-specialist variant

Four measured design findings behind this variant. The router and the specialists want *opposite* feature sets — the 273-column set routes at 0.9703 versus 0.9161 for the

81-column set, but collapses the depression specialist to 0.5280, below its own majority, because 273 dimensions against 53 subjects is starvation; using the rich set for stage 1 and the compact set for stage 2 was worth +0.033, the single largest gain in the track. The guard is load-bearing and must be decided on **inner** cross-validation, since deciding it on outer-fold accuracy would be test-set selection dressed as model selection. Hyperparameter tuning was worth +0.0002. And letting the model select its own feature view per fold made results *worse* by 0.025, because three-fold inner CV over ~40 subjects cannot distinguish the candidate views and the selection thrashes.

4.2 Design

4.2.1 Data Flow Diagram

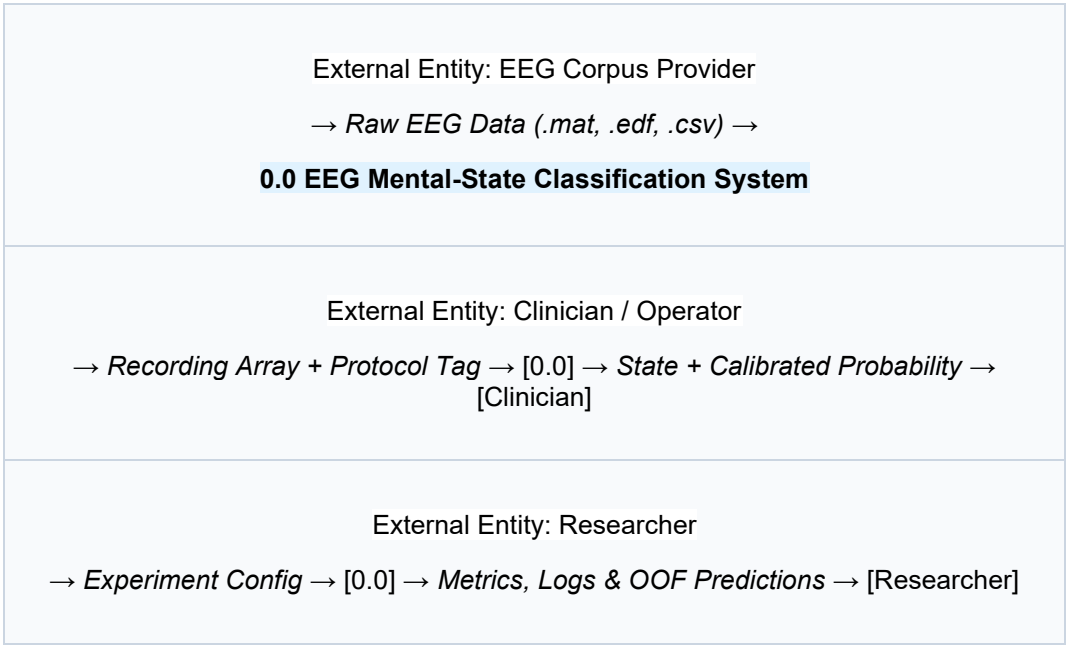


Fig. 4. Level-0 data flow diagram (context diagram)

1.0 Ingest & Parse Identity: Reads raw corpora, strictly extracts verified subject IDs → Emits harmonised arrays.

2.0 Harmonise Channels & Rate: Projects to canonical 6 frontal electrodes (Fp1, Fp2, F7, F3, F4, F8) & resamples to 128 Hz.

3.0 Preprocess, Window & Gate: 0.5–45 Hz bandpass + notch, global z-score, 4s windowing, k=10.0 variance gating → Stores in **D1: Combined Dataset Store**.

4.0 Feature Extraction: Extracts v1 81-col, v2 273-col, and Riemannian tangent

covariances → Stores in **D2: Feature Store**.
5.0 Train & Evaluate (Subject-Grouped CV): Executes 5-fold grouped CV, trains protocol-gated and hierarchical models → Saves to **D3: Results Store** and **D4: Model Registry**.
6.0 Protocol-Gated Inference: Loads fitted estimators from D4, accepts new raw recording + protocol tag, validates input, executes inference → Emits prediction to Clinician.

Fig. 5. Level-1 data flow diagram

Explanation. Process 1.0 is where subject identity is established, and it is security-critical for the evaluation rather than for access control: a subject retained under two identities silently defeats grouped cross-validation. Process 3.0 writes the single canonical dataset that every downstream track reads, which is why any change to labels, channel maps or windowing requires a full rebuild and an explicit cache invalidation — the feature cache is keyed only by filename and would otherwise be reused silently against new data. Process 5.0 enforces subject-grouped splitting; it is the only process permitted to read labels, and the deployment path (6.0) never touches D1 or D3.

4.2.2 Use Case Diagram

Actor	Supported Use Cases
Data Provider (System)	Provide raw EEG corpora (.mat, .edf, .csv) & acquisition metadata
Researcher	Build unified dataset; Verify data invariants; Extract features (v1, v2, Riemannian); Train models; Evaluate under subject-grouped CV; Measure device-disjoint transfer; Compute baselines & controls; Consolidate results; Serialise deployable model
Clinician / Operator	Record EEG session; Submit recording + protocol tag; Receive predicted state + probability; Review confidence & operating point

Fig. 6. Use case diagram

Actor responsibilities. The *Researcher* owns everything from dataset construction to

model serialisation. The *Clinician/Operator* interacts only with the deployment path and, crucially, supplies the protocol tag — the routing bit the architecture depends on. The *Data Provider* is the external corpus repository. Note that Evaluate and Measure transfer both include Compute baselines & controls: reporting a metric without its baseline is not a permitted path through the system.

4.2.3 Class Diagram

CorpusRegistry + CORPORA: dict + VAR_GATE_K: float = 10.0 + CANONICAL_CHANNELS: list + get(corpus_id): CorpusSpec + validate_contract(): bool	**Preprocessor** + project_channels(x): ndarray + resample(x, fs): ndarray + bandpass_notch(x): ndarray + normalize_global(x): ndarray + window(x): ndarray + variance_gate(x, k): ndarray
Dataset + X: ndarray (N, 6, 512) + y_depression: ndarray (N,) + y_stress: ndarray (N,) + subjects: ndarray (N,) + corpus: ndarray (N,) + verify(): bool	**FeatureExtractor & Derivatives** + transform(X): ndarray • HandcraftedV1 (81-col) • HandcraftedV2 (273-col) • RiemannianTangent (recenter, project)
ProtocolGatedModel + backbone: EEGNet + depression_head: BinaryHead + stress_head: BinaryHead + theta_dep, theta_str: float + masked_bce(y_true, y_pred): float + predict(x, protocol): str	**Evaluator & Splitter** + SubjectGroupedSplitter (greedy, deterministic) + tune_threshold(y, p): float + score_with_tuned_threshold(): dict + permutation_null(): float + device_shortcut_baseline(): float

Fig. 7. Class diagram

Design notes. CorpusRegistry is the only place a dataset path, native sampling rate or channel index is written down, satisfying NFR8 — a corpus is corrected in one location. SubjectGroupedSplitter exposes assertion methods rather than trusting the library, because the split is the single most consequential component in the system. Predictor deliberately *reuses* Preprocessor rather than reimplementing it: the normalisation is global rather than per-channel, and a divergent reimplementation would silently corrupt every prediction.

4.2.4 Sequence Diagram

Step	Execution Flow & Messages
1–4	**Dataset Construction:** Researcher triggers build() → DatasetBuilder invokes CorpusLoader across 6 corpora → extracts subject IDs → Preprocessor executes projection, filtering, global z-scoring, windowing → Persists 15,728 segments across 108 subjects to DatasetStore.
5–7	**Feature Transformation:** FeatureExtractor computes v1 (81), v2 (273), and tangent-space covariance matrices → Stores in feature store.
8–13	**5-Fold Grouped Training & Threshold Tuning:** SubjectGroupedSplitter generates subject-disjoint folds → Evaluates ProtocolGatedModel with masked BCE loss → Tunes θ on validation balanced accuracy → Transfers threshold as rank quantile to test fold.
14–16	**Baselines & Result Persistence:** Computes majority baseline, device shortcut baseline (0.6267), permutation null → Writes JSON artefacts and OOF predictions to ResultStore.
17–20	**Operational Deployment:** Clinician submits raw recording + protocol tag → Predictor validates shape and canonical channels → Preprocesses and windows → Dispatches to gated binary head → Emits aggregated prediction and probability.

Fig. 8. Sequence diagram — training and inference

Reading the sequence. Three interactions carry most of the design intent. Step 3 (parse_subject_id) is where grouped evaluation is either preserved or silently destroyed. The assert_no_leakage() call is unconditional rather than a debug option. And in the deployment block, threshold application happens *after* per-window aggregation rather than per window, because window-level votes within a subject are strongly correlated and aggregating decisions rather than scores discards the ranking information that ROC-AUC measures.

5. REFERENCES

Journals — IEEE Format

[1] R. J. Davidson, "Anterior electrophysiological asymmetries, emotion, and depression: Conceptual and methodological conundrums," *Psychophysiology*, Vol. 35, No. 5, (1998), pp. 607–614.

[2] J. B. Henriques and R. J. Davidson, "Left frontal hypoactivation in depression," *Journal of Abnormal Psychology*, Vol. 100, No. 4, (1991), pp. 535–545.

[3] W. Klimesch, "EEG alpha and theta oscillations reflect cognitive and memory performance: A review and analysis," *Brain Research Reviews*, Vol. 29, No. 2–3, (1999), pp. 169–195.

[4] O. Jensen and C. D. Tesche, "Frontal theta activity in humans increases with memory load in a working memory task," *European Journal of Neuroscience*, Vol. 15, No. 8, (2002), pp. 1395–1399.

[5] T. Donoghue, M. Haller, E. J. Peterson, P. Varma, P. Sebastian, R. Gao, T. Noto, A. H. Lara, J. D. Wallis, R. T. Knight, A. Shestyuk, and B. Voytek, "Parameterizing neural power spectra into periodic and aperiodic components," *Nature Neuroscience*, Vol. 23, No. 12, (2020), pp. 1655–1665.

[6] H. Cai, Y. Gao, S. Sun, N. Li, F. Tian, H. Xiao, J. Li, Z. Yang, X. Li, Q. Zhao, Z. Liu, Z. Yao, M. Yang, H. Peng, J. Zhu, X. Zhang, G. Gao, F. Zheng, R. Li, Z. Guo, R. Ma, J. Yang, L. Zhang, X. Hu, Y. Li, and B. Hu, "MODMA dataset: A multi-modal open dataset for mental-disorder analysis," *Scientific Data*, Vol. 9, (2022), Art. no. 178.

[7] W. Mumtaz, S. S. A. Ali, M. A. M. Yasin, and A. S. Malik, "A machine learning framework involving EEG-based functional connectivity to diagnose major depressive disorder (MDD)," *Medical & Biological Engineering & Computing*, Vol. 56, No. 2, (2018), pp. 233–246.

[8] R. Ghosh, N. Deb, K. Sengupta, A. Phukan, N. Choudhury, S. Kashyap, S. Phadikar, R. Saha, P. Das, N. Sinha, and P. Dutta, "SAM 40: Dataset of 40 subject EEG recordings to monitor the induced-stress while performing Stroop colour-word test, arithmetic task, and mirror image recognition task," *Data in Brief*, Vol. 40, (2022), Art. no. 107772.

[9] Cognitive Load EEG Dataset Contributors, "EEG dataset for cognitive load estimation under multi-level task difficulty," *Data in Brief*, (2025). *(Author list and article number to be completed from publisher page)*

[10] I. Zyma, S. Tukaev, I. Seleznov, K. Kiyono, A. Popov, M. Chernykh, and O.

Shpenkov, "Electroencephalograms during mental arithmetic task performance," *Data*, Vol. 4, No. 1, (2019), Art. no. 14.

[11] J. A. Miranda-Correa, M. K. Abadi, N. Sebe, and I. Patras, "AMIGOS: A dataset for affect, personality and mood research on individuals and groups," *IEEE Transactions on Affective Computing*, Vol. 12, No. 2, (2021), pp. 479–493.

[12] S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, and I. Patras, "DEAP: A database for emotion analysis using physiological signals," *IEEE Transactions on Affective Computing*, Vol. 3, No. 1, (2012), pp. 18–31.

[13] S. M. Park, B. Jeong, D. Y. Oh, C.-H. Choi, H. Y. Jung, J.-Y. Lee, D. Lee, and J.-S. Choi, "Identification of major psychiatric disorders from resting-state electroencephalography using a machine learning approach," *Frontiers in Psychiatry*, Vol. 12, (2021), Art. no. 707581.

[15] A. Barachant, S. Bonnet, M. Congedo, and C. Jutten, "Multiclass brain–computer interface classification by Riemannian geometry," *IEEE Transactions on Biomedical Engineering*, Vol. 59, No. 4, (2012), pp. 920–928.

[16] M. Congedo, A. Barachant, and R. Bhatia, "Riemannian geometry for EEG-based brain–computer interfaces: A primer and a review," *Brain-Computer Interfaces*, Vol. 4, No. 3, (2017), pp. 155–174.

[17] P. Zanini, M. Congedo, C. Jutten, S. Said, and Y. Berthoumieu, "Transfer learning: A Riemannian geometry framework with applications to brain–computer interfaces," *IEEE Transactions on Biomedical Engineering*, Vol. 65, No. 5, (2018), pp. 1107–1116.

[18] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, and F. Yger, "A review of classification algorithms for EEG-based brain–computer interfaces: A 10 year update," *Journal of Neural Engineering*, Vol. 15, No. 3, (2018), Art. no. 031005.

[19] B. Hosseinifard, M. H. Moradi, and R. Rostami, "Classifying depression patients and normal subjects using machine learning techniques and nonlinear features from EEG signal," *Computer Methods and Programs in Biomedicine*, Vol. 109, No. 3, (2013), pp. 339–345.

[21] M. M. R. Khan et al., "EEG-based mental stress detection using support vector machine and ensemble learning with optimised preprocessing," (2024), reported accuracy 96.85%. *(Full venue details to be completed by author)*

[22] S. K. Yadav et al., "Multimodal audio–EEG stress assessment using RELIEFF feature selection with KNN and Naive Bayes classifiers," (2023), reported accuracy

88%. *(Full venue details to be completed by author)*

[23] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces," *Journal of Neural Engineering*, Vol. 15, No. 5, (2018), Art. no. 056013.

[24] Y. Ding, N. Robinson, S. Zhang, Q. Zeng, and C. Guan, "TSception: Capturing temporal dynamics and spatial asymmetry from EEG for emotion recognition," *IEEE Transactions on Affective Computing*, Vol. 14, No. 3, (2023), pp. 2238–2250.

[25] Y. Song, Q. Zheng, B. Liu, and X. Gao, "EEG Conformer: Convolutional transformer for EEG decoding and visualization," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, Vol. 31, (2023), pp. 710–719.

[26] B. Ay, O. Yildirim, M. Talo, U. B. Baloglu, G. Aydin, S. D. Puthankattil, and U. R. Acharya, "Automated depression detection using deep representation and sequence learning with EEG signals," *Journal of Medical Systems*, Vol. 43, No. 7, (2019), Art. no. 205.

[27] A. Seal, R. Bajpai, J. Agnihotri, A. Yazidi, E. Herrera-Viedma, and O. Krejcar, "DeprNet: A deep convolution neural network framework for detecting depression using EEG," *IEEE Transactions on Instrumentation and Measurement*, Vol. 70, (2021), pp. 1–13.

[31] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, "Domain-adversarial training of neural networks," *Journal of Machine Learning Research*, Vol. 17, No. 59, (2016), pp. 1–35.

[32] O. Özdenizci, Y. Wang, T. Koike-Akino, and D. Erdoğmuş, "Learning invariant representations from EEG via adversarial inference," *IEEE Access*, Vol. 8, (2020), pp. 27074–27085.

[33] H. He and D. Wu, "Transfer learning for brain–computer interfaces: A Euclidean space data alignment approach," *IEEE Transactions on Biomedical Engineering*, Vol. 67, No. 2, (2020), pp. 399–410.

[34] G. Varoquaux, "Cross-validation failure: Small sample sizes lead to large error bars," *NeuroImage*, Vol. 180, (2018), pp. 68–77.

[35] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, Vol. 6, No. 4, (2012), pp. 1–21.

[36] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, Vol. 4, No. 9, (2023), Art. no. 100804.

[37] R. T. Schirrmeister, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggensperger, M. Tangermann, F. Hutter, W. Burgard, and T. Ball, "Deep learning with convolutional neural networks for EEG decoding and visualization," *Human Brain Mapping*, Vol. 38, No. 11, (2017), pp. 5391–5420.

[38] S. T. Noor et al., "Depression risk prediction from physiological signals using an LSTM autoencoder with a recurrent network," (2023), reported accuracy 97%. *(Full venue details to be completed)*

[39] S. A. Bakry et al., "Computer-aided diagnosis of depression from multimodal data: A review," (2023). *(Full venue details to be completed)*

[40] V. Darbar et al., "Deep neural network based classification of mental states from EEG," (2024). *(Full venue details to be completed)*

[41] N. Milakhina et al., "Delta and gamma spectral markers of adaptation in resting-state EEG of Siberian migrants," (2020). *(Full venue details to be completed)*

[42] U. P. Shukla et al., "Multichannel EEG DeepNet for depression screening." *(Year and venue to be completed)*

[43] S. R. Chowdhury et al., "Preclinical validation of EEG and tDCS electrodes in a rat model," (2023). *(Full venue details to be completed)*

Conferences — IEEE Format

[14] R.-N. Duan, J.-Y. Zhu, and B.-L. Lu, "Differential entropy feature for EEG-based emotion classification," In *2013 6th International IEEE/EMBS Conference on Neural Engineering (NER)*, pp. 81–84, IEEE, 2013.

[20] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 785–794, ACM, 2016.

[28] J. Wang, S. Zhao, Z. Luo, Y. Zhou, H. Jiang, S. Li, T. Li, and G. Pan, "CBraMod: A criss-cross brain foundation model for EEG decoding," In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*, 2025.

[29] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu, "Large brain model for learning generic representations with tremendous EEG data in BCI," In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024.

[30] C. Yang, M. B. Westover, and J. Sun, "BIOT: Biosignal transformer for cross-data learning in the wild," In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36, pp. 78240–78260, 2023.

[44] Md. M. R. Khan et al., "A two-stage hybrid classifier for EEG-based stress detection on SAM-40 using discrete wavelet transform and filter banks," (2025), reported accuracy 95.56%. *(Full venue details to be completed)*

[45] A. Singh et al., "ResKAN: Residual Kolmogorov–Arnold encoder with cross-attention for multimodal depression detection on MODMA," (2026), reported accuracy 99%. *(Full venue details to be completed)*

[46] N. Sakib et al., "Perceived-stress classification from EEG using bagged tree ensembles," (2026). *(Full venue details to be completed)*

Books

[47] S. Sanei and J. A. Chambers, *EEG Signal Processing and Machine Learning*, 2nd ed., Wiley, 2021.

[48] P. L. Nunez and R. Srinivasan, *Electric Fields of the Brain: The Neurophysics of EEG*, 2nd ed., Oxford University Press, 2006.

[49] C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.

[50] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.

Weblinks

[51] MODMA Dataset, Lanzhou University (accessed September 2026).

[52] SAM 40 Dataset, *Data in Brief* (accessed September 2026).

[53] AMIGOS Dataset, Affect, Personality and Mood Research (accessed September 2026).

[54] EEG During Mental Arithmetic Tasks (EEGMAT), PhysioNet (accessed September 2026).

[55] DASPS Database, IEEE DataPort (accessed September 2026).

[56] Cognitive Load EEG Dataset, *Data in Brief* (accessed September 2026).

[57] DREAMER Dataset, Zenodo (accessed September 2026).

[58] WESAD Dataset, University of Siegen (accessed September 2026).

[59] MNE-Python Documentation (accessed September 2026).

[60] pyRiemann Documentation (accessed September 2026).

[61] braindecode Documentation (accessed September 2026).